Experiment-13

Limitations of Low-Code Platforms for a Complex Enterprise Use Case

Aim:

To study and analyze the limitations of low-code development platforms when implementing complex enterprise applications, focusing on scalability, customization, integration, security, and performance.

Theory:

Low-code development platforms allow users to build applications using visual interfaces, drag-and-drop components, predefined templates, and minimal programming. They help reduce development time and simplify application development.

However, complex enterprise applications often require advanced business logic, large-scale data processing, integration with multiple systems, strict security controls, and high performance. Low-code platforms may face limitations in these areas.

The major limitations are:

1. **Limited Customization:** Advanced business requirements may not be supported by built-in components and may require custom code.
2. **Scalability Challenges:** Applications handling large numbers of users and records may require additional performance optimization and architectural planning.
3. **Integration Limitations:** Connecting legacy systems, external APIs, and multiple enterprise services may require additional development and configuration.
4. **Security and Compliance:** Enterprise applications must meet strict access-control, privacy, auditing, and regulatory requirements.
5. **Performance Issues:** Complex workflows, large datasets, and multiple integrations can increase processing time.
6. **Vendor Lock-in:** Applications built using platform-specific tools and scripting languages may be difficult to migrate to another platform.
7. **Limited Control:** Developers may have less control over the underlying infrastructure, deployment environment, and internal application architecture.

For example, a large enterprise resource planning (ERP) system may require real-time inventory tracking, payroll processing, financial reporting, and integration with several databases. A low-code platform can support some of these functions, but complex requirements may need custom programming, external services, or a hybrid development approach.

Procedure:

1. Select a low-code platform such as Zoho Creator for study.
2. Examine its application development features, including forms, reports, dashboards, and workflows.
3. Consider a complex enterprise use case, such as an ERP system managing inventory, employees, payroll, and financial transactions.
4. Identify the application's requirements for data processing, scalability, security, customization, and external integration.
5. Analyze the platform's built-in capabilities for meeting these requirements.
6. Identify situations where custom code, external APIs, or additional backend services may be required.
7. Compare the advantages of low-code development with its limitations for complex enterprise applications.
8. Record the findings and determine whether a low-code, pro-code, or hybrid approach is most suitable.

Result:

Thus, the limitations of low-code platforms for complex enterprise applications were studied and analyzed. The study identified challenges related to customization, scalability, integration, security, performance, and vendor lock-in. It was concluded that low-code platforms are suitable for rapid application development, while complex enterprise requirements may benefit from a combination of low-code tools and conventional programming